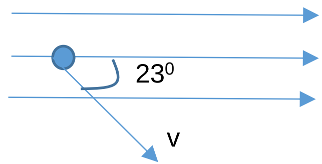
25 **Answer: A**

Conventional current direction due to electron going downwards at P

From Fleming's Left Hand Rule, electromagnetic force, F on electron is into the plane



B field (0.084 T)

F will always be perpendicular to the component of velocity normal to the B field. Hence electron will move in circular motion. At the same time, the component of velocity parallel to the B field remains constant. Hence, the electron will move in a helix.

F = Bqv sin θ

$$v = \frac{7.3 \times 10^{-16}}{(0.084)(1.6 \times 10^{-19}) \sin 23^0}$$

$= 1.4 \times 10^5 \text{ ms}^{-1}$.

26 **Answer: C**